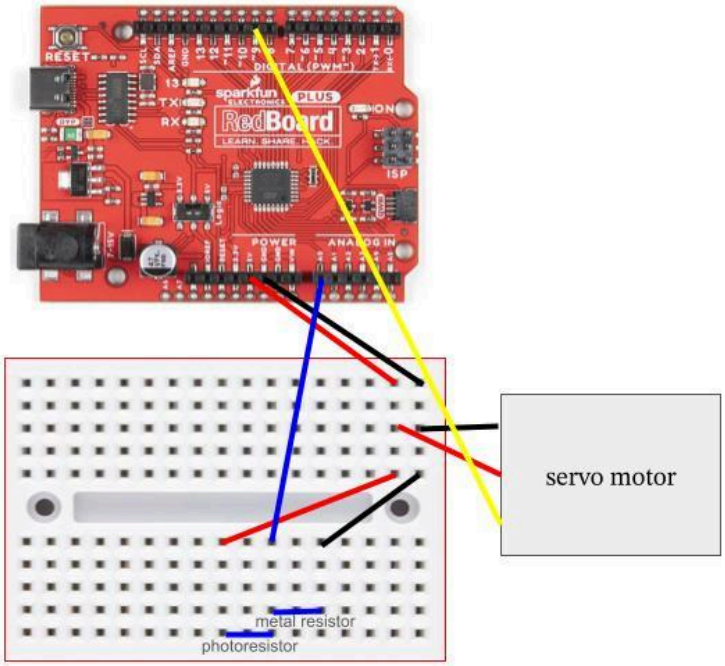
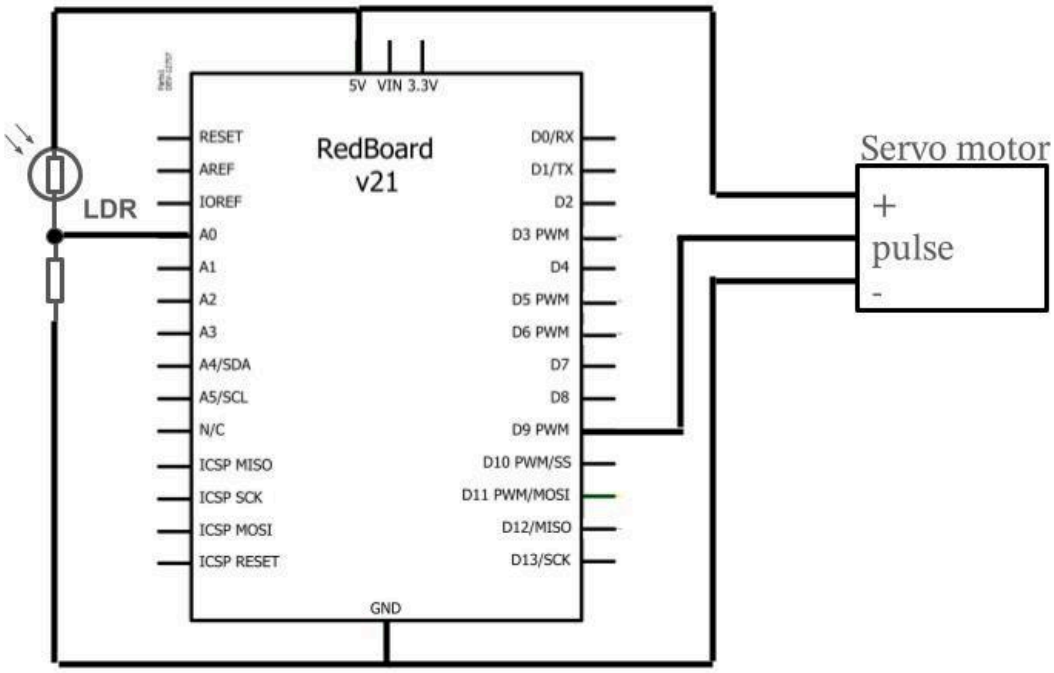
Materials:

- SparkFun RedBoard
- BreadBoard
- Photoresistor
  - 100-200KΩ bright resistance
  - 10MΩ dark resistance
- 470KΩ metal film resistor
- Servo Motor
- 10 wires





```
#include <Servo.h>

Servo Servo1;

int servoPin = 9;
int potPin = A0;
int light_val = 1010;

void setup() {
    Serial.begin(9600);
    Servo1.attach(servoPin);
}


void loop() {
    Serial.print("read: ");
    int reading = analogRead(potPin);
    Serial.println(reading);

    if (reading >= light_val) {
      // light on, open gate
      Servo1.write(180);
      delay(2000); //stability delay
    } else {
      // no light, close gate
      Servo1.write(0);
    }

}
```